

# ALI

## JAVIDAN PAK

### CONTACT INFO

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|-------------|-----------------------------------|
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### ABOUT ME

Physics student interested in theoretical and computational approaches to quantum many-body and condensed matter systems, with experience in numerical many-body methods and a growing interest in machine learning for physics.

### EDUCATION

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| <b>B.Sc. IN PHYSICS</b><br><i>Sharif University of Technology</i>                                                                                                                                                                                                                                                                                                      | <b>2022-2026</b> |
| <ul style="list-style-type: none"><li>• GPA: 18.11/20</li><li>• Ranked in the top 10% of the undergraduate Physics cohort.</li><li>• Bachelor's project: "The Adiabatic Theorem".</li><li>• Selected coursework: Computational Physics with Python, Solid State Physics, Implementation of Quantum Computing, Mathematical Physics I &amp; II, Group Theory.</li></ul> |                  |
| <b>M.Sc. IN PHYSICS</b><br><i>Sharif University of Technology</i><br>(Direct Admission)                                                                                                                                                                                                                                                                                | <b>2026-</b>     |

### RESEARCH & PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                         |                  |
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| <b>UNDERGRADUATE RESEARCHER</b><br><i>Sharif University of Technology</i><br>(The Adiabatic Theorem)                                                                                                                                                                                                                                                                                                    | <b>2025-2026</b> |
| <ul style="list-style-type: none"><li>• Studied adiabatic and nonadiabatic transitions in two-state quantum systems using complex-time methods.</li><li>• Analyzed the time-dependent Schrödinger equation using contour integration, Laurent expansions, and asymptotic analysis.</li><li>• Derived asymptotic transition amplitudes and probabilities in the semiclassical/adiabatic limit.</li></ul> |                  |

**COMPUTATIONAL PHYSICS**  
*Sharif University of Technology*  
(Selected Projects)

2025-2026

- Implemented exact diagonalization of quantum spin chains using tensor products, sparse linear algebra, and Lanczos methods.
- Implemented Matrix Product States and two-site DMRG with SVD-based truncation and variational optimization.
- Compared DMRG with exact diagonalization and studied bond-dimension dependence and entanglement entropy.

**STOCHASTIC PROCESSES**  
*Sharif University of Technology*  
(Course Project)

2026

- Implemented Gaussian Process regression from scratch using RBF and Brownian kernels.
- Developed a Metropolis-Hastings sampler for Bayesian hyperparameter inference and assessed MCMC convergence.
- Simulated Langevin dynamics using Euler-Maruyama discretization and studied convergence to stationary distributions.

**TOPOLOGICAL QUANTUM COMPUTATION**  
*Sharif University of Technology*  
(Course Project)

2025

- Studied Fibonacci and Ising anyon models and their use in fault-tolerant quantum computation.
- Derived fusion rules, F-matrices, and braiding operators for non-Abelian anyons.
- Analyzed topological protection and the conditions for universal quantum computation through anyon braiding.

**PUBLICATIONS**

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**Common (Mis)conceptions of Quantum Mechanics**  
*Takaneh Magazine, Sharif University of Technology*  
(Scientific Team Member)

2026

- Contributed sections on quantum superposition and Schrödinger's cat, and the Heisenberg uncertainty principle.

**SKILLS**

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|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>           | Python, NumPy, SciPy, Matplotlib, SymPy, JAX, Mathematica (basic)                                                              |
| <b>Computational Physics</b> | Monte Carlo, MCMC, numerical ODE/PDE methods, exact diagonalization, sparse linear algebra, Lanczos, tensor networks, MPS/DMRG |
| <b>Machine Learning</b>      | Neural networks, scikit-learn, XGBoost, LightGBM, TensorFlow/Keras                                                             |
| <b>Tools</b>                 | $\text{\LaTeX}$ , Git/GitHub                                                                                                   |